Graphyti Project Summary

The 3D Data Engine for Smart City Robots

Chan Hin Wai Howell

Lee Ern Qi Eunice

Problem and Opportunity

Robotics faces a key bottleneck: a lack of high-quality, spatially structured, safety-critical training data. Urban robots operate in highly complex environments, requiring them to navigate dynamic spaces, interact safely with people, and adapt to unpredictable conditions like weather, obstacles, and changing infrastructure.

The challenge is exacerbated by the fact that datasets are often not transferable across different robotics use cases. For example, aerial drones require top-down perspectives, while ground robots rely on street-level views. Some applications demand dense human data, while others prioritize vehicles, bicycles, or specific edge cases. As a result, there is no one-size-fits-all dataset, and finding or assembling the exact data needed for a specific use case is extremely difficult.

Collecting such data in the real world is expensive, slow, and complex. Teams must capture and label diverse environments under varying conditions before training. As a result, the field is shifting toward synthetic training: using simulation and digital environments to scale data and test systems pre-deployment. Notable examples include Nvidia emphasizing simulation for parallel, physics-based training, and the usage of digital twins for reducing real-world risk [1][2].

This is where Graphyti can create clear value. For example, a company building an outdoor urban delivery robot could use Graphyti to generate a dense city district with different objects and even changing weather. The robot could then be trained on thousands of edge cases: a pavement blocked by construction cones, a pedestrian crossing its path, or low light conditions. These edge cases matter because commercial sidewalk robots like *Starship* will operate under these urban constraints and they need to be well prepared for them [3] [4] .

The broader market also supports this opportunity. *Research and Markets* estimates the synthetic data generation for the robotics market at US$2.48 billion in 2026, projected to reach US$7.71 billion by 2030 at a 32.9% CAGR [6]. At the same time, *MarketsandMarkets* estimates the broader smart robots market will grow from US$16.2 billion in 2025 to US$42.8 billion by 2030 [7]. These figures suggest that both robotics deployment and the supporting infrastructure for training data are growing quickly.

For Graphyti, this creates a practical and timely opportunity: help robotics teams train more safely, iterate more cheaply, and build proprietary datasets without needing to capture every scenario on real city streets, while powering robust systems for urban deployment.

Solution and Value Proposition

Graphyti is a platform that allows companies to create enterprise-scale 3D datasets through a fully no-code workflow. Users can generate virtual worlds from prompts, modify and label environments, and capture training data..

The platform offers three key advantages. First, Graphyti enables scale through procedural generation, allowing companies to create large 3D worlds with near-infinite variations (Fig. 1). This is valuable for robotics, simulation, and urban digital twin models that must generalize across different layouts, conditions, and perspectives.

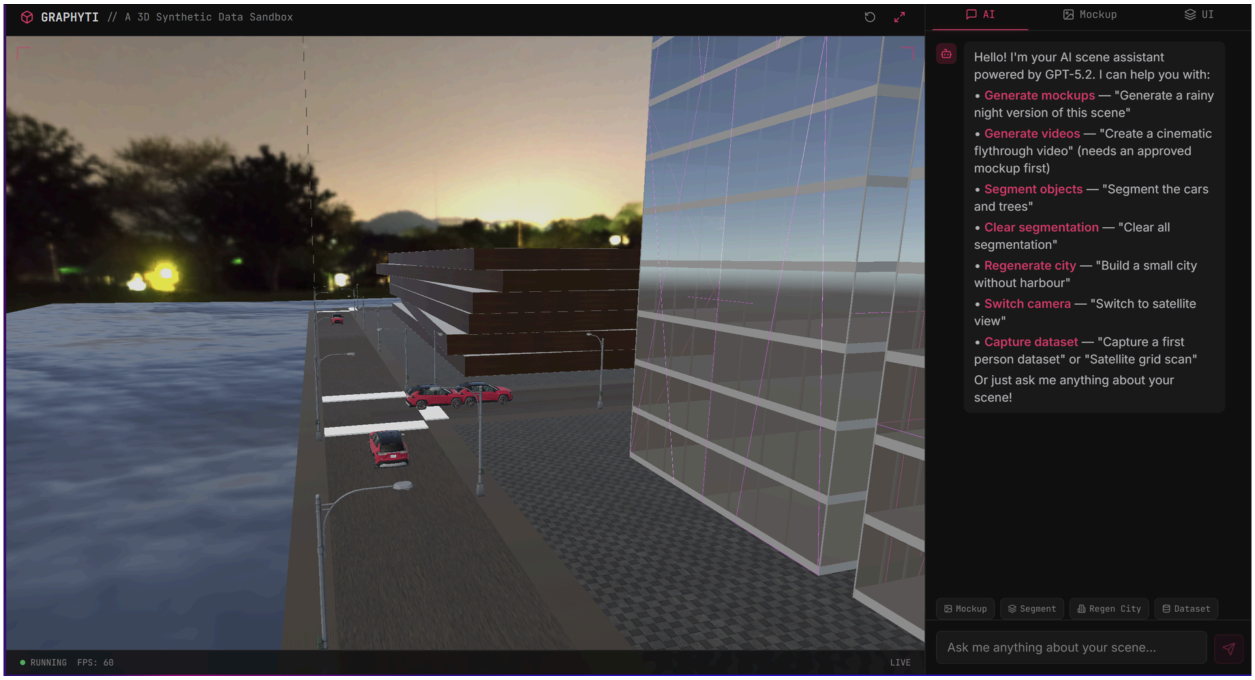

Figure 1: Sample of a city by the sea created via in-house algorithms hosted by Graphyti's web platform

Second, Graphyti offers accessibility with a no-code UI (Fig. 2). Existing synthetic data tools often require coding knowledge or specialist graphics workflows. Graphyti lowers the barrier to creating datasets, eliminating real-world data collection or complex simulation pipelines.

Figure 2: Sample image dataset created on Graphyti's web platform

Third, Graphyti improves controllability. Our datasets are generated from structured 3D environments, customers can train models on scenes with known layouts, object placements, labels, and viewpoints (Fig. 3 & 4). This is especially useful for systems where reliable and predictable outputs matter.

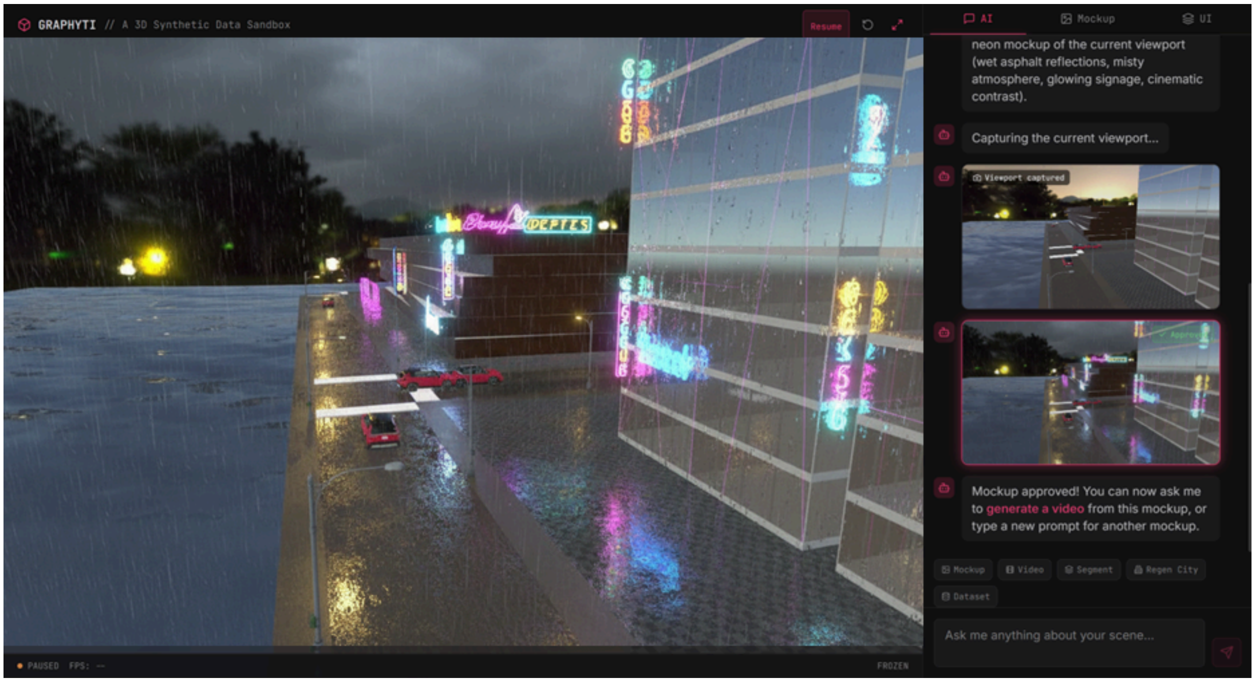

Figure 3: Sample of rain simulation designed on Graphyti's web platform

Figure 4: Simulation of falling tree in a city

Market Potential & Business Model

We estimate that approximately 52,000 companies globally require high-quality 3D data, of which around 16,000 require large-scale 3D datasets. Over the next three years, we intend to target 30 high-potential companies from this larger pool.

Graphyti is a software-based B2B company with a recurring revenue model. Potential customers include robotics companies and computer vision platforms used for urban monitoring and infrastructure management.

We intend to serve customers through three annual pricing tiers: US$240,000, US$100,000, and US$24,000. This structure allows us to serve both enterprise-scale organizations and smaller teams with lower budgets. Based on our target customer mix, we project approximately US$3.4 million in annual recurring revenue by 2028.

Because the product is software-driven and built on reusable procedural systems, it can expand across multiple industries without being rebuilt for each one. Initial distribution will be driven through direct B2B sales, pilot projects, and strategic validation partnerships.

Graphyti can reduce dataset creation costs by up to 80% compared to manual real-world collection, while dramatically reducing turnaround time.

Competitive Position

Graphyti is by far the only platform focused specifically on synthetic data generation through a fully no-code interface. While companies such as Bifrost and Sky Engine AI operate in adjacent areas of 3D synthetic data, they require coding knowledge which increases barriers to entry.

Graphyti's primary competitive advantage lies in its scalable procedural generation capability, accessed through a user-friendly, prompt-based interface. This technology enables the rapid creation of extensive 3D environments, drastically cutting down production time.

Specifically, a team of just three engineers and two artists can build a full-scale 3D world with nearly limitless variations in approximately one week. This represents a time saving of up to 75% compared to the 1–3 months typically required by conventional workflows. Competitors generally need larger teams, longer development cycles, yet still produce fewer variations.

Team, Execution, and Funding

Graphyti is led by a team with expertise in computer graphics research and AI full-stack engineering. Over the past two years, we have developed the project alongside school and have received multiple competition wins (Fig. 5).

Figure 5: Overview of Graphyti's achievements from 2024-2026

We have also validated the product through discussions with World Labs, HTX, NTU, and Zeiss, which have informed both our product direction and commercial strategy.

Our implementation timeline is as follows (Fig. 6):

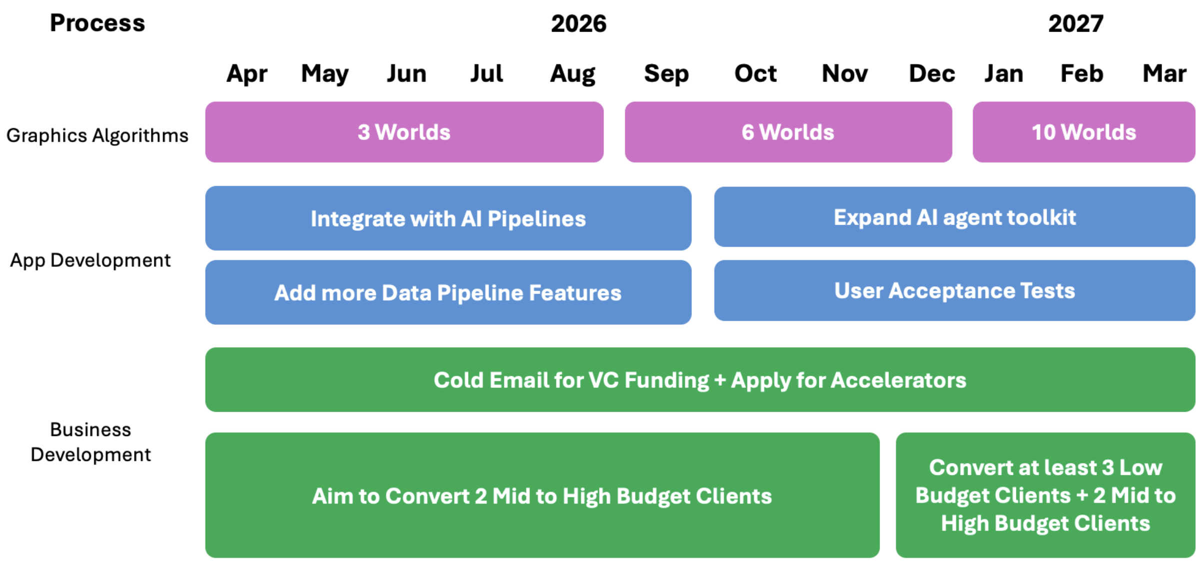

Figure 6: Implementation Timeline 2026-2027

Graphyti intends to build from Singapore. We are seeking capital to expand the team, deepen research and development, and scale cloud deployment infrastructure.

Conclusion

Graphyti addresses a critical need in smart city robotics: the ability to create high-quality, proprietary, and scalable 3D datasets. By replacing slow and expensive real-world collection workflows with controllable synthetic environments, Graphyti enables faster development, stronger data ownership and safer testing.

With a large addressable market, a scalable revenue model, a differentiated product, and a technically capable team, Graphyti is poised to become vital infrastructure for the next generation of smart city systems.

Appendix A: Financial Projections

Metric	2026	2027	2028	2029	2030
High-tier customers	0	2	6	12	20
Mid-tier customers	0	6	15	25	26
Low-tier customers	0	4	9	16	25
Revenue					
ARR - High tier (US$)	$0	$480,000	$1,440,000	$2,880,000	$4,800,000
ARR - Mid tier (US$)	$0	$720,000	$1,800,000	$3,000,000	$3,120,000
ARR - Low tier (US$)	$0	$96,000	$216,000	$384,000	$600,000
Total ARR (US$)	**$0**	**$1,296,000**	**$3,456,000**	**$6,264,000**	**$8,520,000**
YoY ARR growth			166.7%	81.3%	36.0%
Cost of Goods Sold					
COGS % of ARR	0.0%	20.0%	20.0%	20.0%	20.0%
COGS (US$)	$0	$259,200	$691,200	$1,252,800	$1,704,000
Gross profit (US$)	**$0**	**$1,036,800**	**$2,764,800**	**$5,011,200**	**$6,816,000**
Gross margin %		80.0%	80.0%	80.0%	80.0%
Headcount					
Average engineers	0	3	5	6	7
Average artists	0	2	2	2	3
Engineer salary / year (US$)	$60,000	$80,000	$80,000	$80,000	$80,000
Artist salary / year (US$)	$40,000	$40,000	$40,000	$40,000	$40,000
Engineer salary expense (US$)	$0	$240,000	$400,000	$480,000	$560,000
Artist salary expense (US$)	$0	$80,000	$80,000	$80,000	$120,000
Other operating expenses					
Sales & travel (US$)	$0	$40,000	$80,000	$120,000	$160,000
Software / admin / legal (US$)	$0	$35,000	$60,000	$80,000	$100,000
Total operating expenses ex-COGS (US$)	**$0**	**$395,000**	**$620,000**	**$760,000**	**$940,000**
Profitability					
EBITDA / operating profit (US$)	**$0**	**$641,800**	**$2,144,800**	**$4,251,200**	**$5,876,000**
EBITDA margin %		49.5%	62.1%	67.9%	69.0%

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